MILITARY SPECIFICATION

TEST SET, GUIDED MISSILE LAUNCHER AN/AWM-90A

This specification is approved for use by the Naval Air Systems Command, Department of the Navy and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, and acceptance requirements for the Guided Missile Launcher Test Set AN/AWM-90A. The test set is a microprocessor-controlled stimuli, load, and measurement system programmed to test AERO 5B, LAU-118(V)1/A, LAU-7/A-5, LAU-115/A, LAU-116/A, and LAU-117A(V)2/A Guided Missile Launchers.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards and handbooks of the issue listed in that issue of Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

Federal

PPP-T-60	Tape, Packaging, Waterproof
PPP-T-76	Tape, Pressure Sensitive Adhesive, Paper (For Carton Sealing)
PPP-B-585	Boxes, Wood, Wire-bound
PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiberboard
QQ-S-781	Strapping Steel, and Seals

Military

MIL-P-116	Preservation, Methods of
MIL-C-3643	Connector, Plug, Electrical Series HN, Type UG-59E/U
MIL-C-3650	Connector, Coaxial Radio Frequency, Series LC
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of
MIL-T-18303	Test Procedures; Preproduction and Inspection, for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronics, Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment
MIL-S-19500	Semiconductor Device, General Specification for
MIL-T-28800	Test Equipment for use with Electrical and Electronic Equipment, General Specification for

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MIL-M-38510	Microcircuit, General Specification for
MIL-C-38999	Connector, Electrical, Circular, Miniature, High Density, Quick Disconnect (Bayonet, Threaded, and Breech Coupling), Environment Resistant, Removable Crimp and Hermetic Solder Contacts, General Specification for
MIL-C-39012	Connector, Coaxial, Radio Frequency, General Specification for
MIL-H-46855	Human Engineering Requirements for Military Systems, Equipment and Facilities
MIL-C-55302	Connector, Printed Circuit Subassembly and Accessories
MIL-C-55544	Connector, Electrical, Environment Resistant, for use with Flexible Flat Conductor Cable and Round Wire, General Specification for
MIL-C-83723	Connector, Electrical (Circular, Environment Resistant), Receptacles and Plugs, General Specification for

STANDARDS

Military

MIL-STD-108	Definition of Basic Equipment for Enclosure for Electric and Electronic Equipment
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-147	Palletized Unit Loads Validated October 1974
MIL-STD-198	Capacitor, Selection and Use of
MIL-STD-199	Resistor, Selection and Use of
MIL-STD-454	Standard Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference

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MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-470	Maintainability Program Requirements for Systems and Equipments Prediction
MIL-STD-471	Maintainability Demonstrations
MIL-STD-701	List of Standard Semiconductor Devices
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-810	Environmental Test Methods
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-1188	Commercial Packaging of Supplies and Equipments
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-1562	Lists of Standard Microcircuits

HANDBOOKS

Military

MIL-HDBK-472	Military Standardization Handbook, Maintainability Predictions

Naval Air Systems Command

NAVAIR 16-30AWM90-1	Technical Manual, Intermediate Instruction with Illustrated Parts Breakdown, Guided Missile Launcher Test Set AN/AWM-90A

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

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BULLETINS

ANA Bulletin No. 400	Air Force-Navy Aeronautical Bulletin Electronic Equipment; Aircraft and Guided Missiles, Applicable Documents

DRAWINGS

Naval Air Systems Command

1271AS1000	Test Set, Guided Missile Launcher AN/AWM-90A

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Item description. The AN/AWM-90A test set (see figure 1) shall consist of a self-test adapter for use in self tests, two test cables for performing self test and equipment tests, 13 adapters, for adapting to the various equipments, two power cables for DC and 400 cycle supply, a LAU-7 Detent Wrench, an accessory case, and an instrument case.

3.2 First article. When specified (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.3 Materials, parts, processes. Materials, parts, and processes used in the test set shall be as specified in MIL-T-28800, MIL-STD-454 and 3.3.1 through 3.3.2.2.

3.3.1 Parts. All parts shall be selected from those parts included in ANA Bulletin No. 400, unless the parts listed therein will not perform as required in the intended applications and environments specified herein. If the latter is the case, request for use of nonstandard parts shall be obtained by the contractor in accordance with Requirement 23.

3.3.1.1 Solid state design. All microelectronic devices shall be selected as specified in Requirement 64 of MIL-STD-454 and MIL-STD-883. Hybrid microcircuits, including radio frequency, memory, microwave, and multimeter types are considered microelectronic devices and shall be controlled by Requirement 64 of MIL-STD-454. Hybrid and complex monolithic microcircuits are considered critical items and shall be treated as specified in MIL-STD-785. Any deviation from all solid state construction shall be authorized by the procuring activity prior to initiation.

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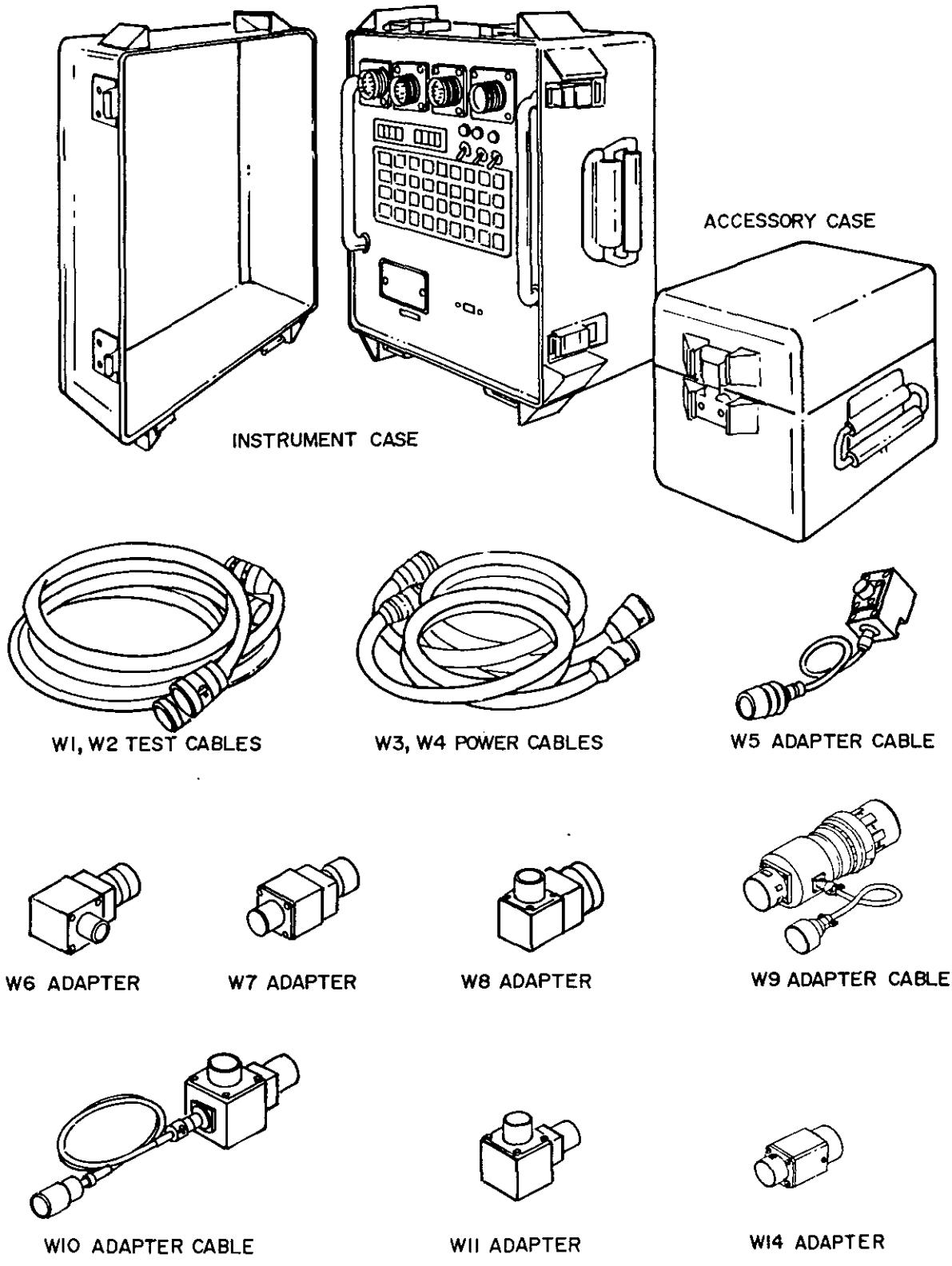
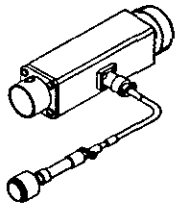
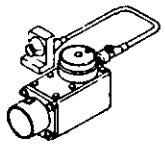


FIGURE 1. Guided Missile Launcher Test Set AN/AWM-90A (Sheet 1 of 2)

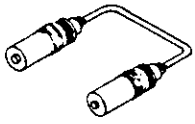
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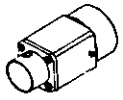
W15 ADAPTER CABLE



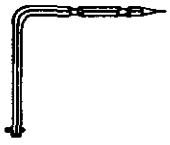
W16 ADAPTER CABLE



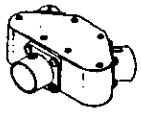
W17 CARTRIDGE ADAPTER



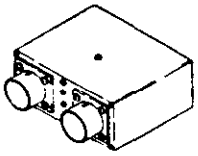
W18 ADAPTER



DETENT WRENCH



W19 ADAPTER



W23 ADAPTER

FIGURE 1. Guided Missile Launcher Test Set AN/AWM-90A (Sheet 2 of 2)

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3.3.1.2 Microcircuits and semiconductors. All microcircuits and semiconductors shall be packaged in hermetic packages. No plastic encapsulated devices shall be used.

3.3.1.2.1 Microcircuit devices. Only microcircuits listed in MIL-STD-1562 and procured as specified in MIL-M-38510 are standard for new design. When nonstandard microcircuit devices are approved for use as specified in the general requirement of MIL-M-38510, Class B shall apply. Nonstandard devices shall be screened and qualified as specified in MIL-STD-883 (Test Methods 5004.3 and 5005.3). The qualification as specified in the requirements of Methods 5005.3, Group C, shall be satisfied by submission of approved test results for all standard and nonstandard microcircuit devices.

3.3.1.2.2 Semiconductor devices. Only JANTX semiconductor devices selected as specified in MIL-STD-701 are standard. When a JANTX device is not listed, the selection of nonstandard devices shall conform to the following order of precedences:

a. A JAN device listed as specified in MIL-STD-701.

b. A JAN device covered as specified by MIL-S-19500 but not listed in MIL-STD-701.

c. A commercial device.

The definition of JANTX, JAN, and TX shall be specified by MIL-S-19500. As a minimum, a TX burn-in requirement as specified in Requirement 30 of MIL-STD-454 shall be required for all nonstandard devices. All semiconductors shall be selected as specified in Requirement 30 of MIL-STD-454. Only solid glass metallurgically bonded axial lead diodes and rectifiers shall be used. When TO-5 and TO-18 packages are required, they shall be limited to the solid metal header type. All semiconductor device junctions shall be protected and no organic or desiccant materials shall be included in the package. Thermocompression wedge bonding shall not be used with aluminum wire. Aluminum TO-3 packages shall not be used.

3.3.1.3 Electrical connectors. Unless otherwise specified in the contract (see 6.2.1), Requirement 10 of MIL-STD-454 shall be applicable except that only MIL-C-38999 and MIL-C-83723 covering circular connectors, and MIL-C-55302 covering printed circuit board connectors shall be employed. Connectors for use with flat cable shall meet the specified requirements of MIL-C-55544. RF coaxial connectors shall be as specified in MIL-C-39012, Series TNC, SC, SMA, or SMB, except where the power or voltage rating of SC series does not meet application requirements. In these cases, Series HN as specified in MIL-C-3643 or Series LC as specified in MIL-C-3650 shall be used.

3.3.1.4 Capacitors and resistors. Only established reliability capacitors and resistors as specified in MIL-STD-198 and MIL-STD-199 are standard.

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3.3.2 Materials.

3.3.2.1 Moisture and fungus resistance. The test set shall be designed with materials that are moisture resistant and are not nutrient to fungi. The use of nutrient material is acceptable in hermetically sealed assemblies, paper capacitors, and transformers. Fungus inert materials are specified in MIL-STD-454, Requirement 4. Other materials shall be certified fungus-resistant or shall be tested as specified in Method 508.1 of MIL-STD-810. The parts used in the design shall be moisture resistant. Absorption, retention, or release of moisture by a part shall not degrade or affect performance.

3.3.2.2 Corrosion. Corrosion prevention and control as an overall concept shall be emphasized during design and fabrication. The test set shall be designed to prevent corrosion that would adversely affect its functions during the specified service life consistent with required reliability and maintenance concepts. Requirements 15 and 16 of MIL-STD-454 shall apply.

3.3.3 Interchangeability. The test set shall meet the interchangeability requirements of MIL-T-28800. Drawing 1271AS1000 shall be standard for measuring interchangeability of all applicable units, assemblies, subassemblies, and parts.

3.4 Design and construction. The test set shall be manufactured in accordance with Drawing 1271AS1000, except that errors of omission or commission on the drawing may be incompatible with the requirements of this specification. In any case, it is a condition of final acceptance that the equipment manufactured shall meet all the design, construction, and workmanship requirements of this specification and applicable test documents.

3.4.1 Reliability.

3.4.1.1 Operational stability. The test set shall operate with required performance, continuously or intermittently for a period of at least 1000 hours. During this time, it shall not be necessary to readjust any controls that are inaccessible to the operator during normal use.

3.4.1.2 Operating life. The test set shall have a minimum total operating life of 10000 hours with minimal servicing and replacement of parts.

3.4.1.3 Reliability in mean-time-between-failures (MTBF). The test set shall have a specified mean (operating) time between failures of 800 hours when tested in accordance with 4.5.3.

3.4.2 Maintainability. Maintainability provisions shall be incorporated within the design to provide for service, access and repair of all assemblies. The replacement of major components within the equipment shall be accomplished without the need for special tools. Normal routine testing of the test set shall be accomplished using the standard test equipment normally available in Intermediate Maintenance Shops.

3.4.2.1 Mean-time-to-repair (MTTR). The mean-time-to-repair (MTTR) of the equipment shall be not greater than two hours when replacing circuit modules, plug-ins, assemblies and parts that are intended for replacement at the intermediate level. Only where design changes shall cause significant

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deviations from procuring activity drawings, a prediction of MTTR shall be made in accordance with MIL-STD-470 and MIL-HDBK-472, Procedure II, and shall be demonstrated in accordance with 4.5.3.

3.4.2.2 Built-in-test (BIT). The BIT shall perform self tests to verify proper operation of the test set. The BIT modes shall be as follows:

a. Limited turn-on BIT.

b. Full turn-on BIT.

c. Self test.

3.4.2.2.1 Limited turn-on BIT. The BIT Limited Turn-On mode shall perform automatically when test set is energized, or when RESET is pressed and there is a launcher adapter or adapter cable connected. In this test mode, the memory, microprocessor, digital latches, measurement system, and internal and external power supplies shall be functionally tested.

3.4.2.2.2 Full turn-on BIT. The BIT Full Turn-On mode shall perform automatically when the test set is energized or RESET is pressed and no adapter cables are connected. In this test mode, the analog stimuli circuits are functionally tested in addition to circuits tested in the BIT Limited Turn-On mode.

3.4.2.2.3 Self test. The Self Test shall verify test cable integrity, performance of internal loads and the remainder of the stimuli circuitry using the self test adapter (see figure 1). Functions/circuits tested in this mode are in addition to the areas tested in BIT Full Turn-On mode. The self test shall be capable of performing its test functions both automatically and manually. Automatic test shall run the entire self test while manual test shall step through the test one step at a time. (This test does not check the launcher adapters.)

3.4.2.3 Lamp test. The test set shall be capable of testing all lamps in the keyboard and all digits in the display by illuminating each bank of the keyboard lamps for a few seconds starting with the left most bank, then advancing to the next bank and finally turning on all the lamps in the display.

3.4.2.4 Keyboard switch test. This test shall be capable of verifying the condition of all keyswitches in the keyboard. The keys shall illuminate one at a time and, when pressed, either turn off if good or stay on if bad.

3.4.3 Transportability. The test set shall be transportable by air, rail, truck, and ship when packaged as specified in section 6.

3.4.4 Standard conditions. The following conditions shall be used to establish the normal performance characteristics under standard conditions for performing laboratory bench tests:

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a.	Temperature	Room ambient (30 ± 10°C)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative humidity
e.	Input Power	115 ± 11.5 volts, 400 ± 20 Hz, 3Ø at 12 amperes. +28 +2 -6 volts dc at 20 amperes

3.4.4.1 Service conditions. The test set shall operate as required under any of the environmental service conditions or combination of these conditions as specified in MIL-T-28800 for Class 3 equipment. As a minimum, the test set shall be required to meet specified performance while being exposed to the following environmental conditions.

3.4.4.1.1 High temperature. The test set shall meet the high temperature requirements of MIL-STD-810, Method 501, Procedure I as modified by MIL-T-28800.

3.4.4.1.2 Low temperature. The test set shall meet the requirements of MIL-STD-810, Method 502, Procedure I, as modified by MIL-T-28800.

3.4.4.1.3 Humidity. The test set shall operate as required after being subjected to a humidity test in accordance with MIL-STD-810, Method 507, Procedure III. The equipment shall not be damaged as a result of the humidity test specified herein.

3.4.4.1.4 Watertight. The test set shall meet specified performance and accuracy requirements after being exposed to a watertight test in accordance with MIL-STD-108 (see 4.7.4).

3.4.4.1.5 Altitude. The test set shall meet specified performance and accuracy requirements during and after being subjected to altitude test in accordance with MIL-STD-810, Method 500, Procedure I, except the maximum simulated altitudes shall be 40,000 feet (nonoperating) and 15,000 feet (operating).

3.4.4.1.6 Vibration. The test set shall meet the specified performance and accuracy requirements during and after being subjected to vibration test in accordance with MIL-STD-810, Method 514, Procedure X modified by 4.7.6.

3.4.4.1.7 Shock. The test set shall meet specified performance and accuracy requirements after being subjected to shock test in accordance with MIL-STD-810, Method 516, Procedure I and 4.7.7.

3.4.4.1.7.1 Transit drop. The test set shall meet specified performance and accuracy requirements after being subjected to drop test in accordance with MIL-STD-810, Method 516, Procedure II and 4.7.8.

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3.4.4.1.7.2 Bench handling. The test set shall meet specified performance and accuracy requirements after being subjected to bench handling test in accordance with MIL-STD-810, Method 516, Procedure V as modified in 4.7.9.

3.4.4.2 Warm-up time. The test set shall comply with specified performance requirements after a 10-minute warm-up period following a two hour nonoperating temperature stabilization period or as specified in the applicable test method.

3.4.4.3 Input power. The test set shall be capable of specified operating when supplied input power that meets the requirements of MIL-T-28800. The power required shall be of the following types and shall not exceed the following:

a. AC Power, Three Phase, 115 ± 11.5 Volt, 400 ± 20 Hz, 12 VA

b. +28 +2, -6 Volts DC at 20 ampere capacity

3.4.4.3.1 Protection. The test set shall be capable of withstanding input variance as imposed by abnormal electrical system operations as specified in MIL-STD-704.

3.4.4.4 Electromagnetic interference. The generation of electromagnetic interference by the test set and the susceptibility of the test set to electromagnetic interference shall be in accordance with MIL-STD-461B as referenced in MIL-T-28800.

3.4.4.4.1 Conducted emissions, power and interconnecting leads. Conducted emissions from the test set power and interconnecting leads shall be within the limits of MIL-STD-461B, Test Method CE03, over the frequency range of 0.015 to 50 MHz.

3.4.4.4.2 Conducted emissions, power leads, spikes, time domain. Conducted emission spikes and switching transients measured on the power leads in accordance with MIL-STD-462 shall not exceed the limits of MIL-STD-461B, Test Method CE07.

3.4.4.4.3 Conducted susceptibility, power leads. The test set shall not be susceptible to signals of 30 Hz to 400 MHz injected on the dc power leads at rms voltage levels of MIL-STD-461B, Test Methods CS01 and CS02.

3.4.4.4.4 Conducted susceptibility, spikes, power leads. The test set shall not be susceptible to spike interference injected on the dc power leads in accordance with MIL-STD-461B, Test Method CS06.

3.4.4.4.5 Radiated emissions, electric fields, 14 KHz to 10 GHz. Radiated emission from the test set shall be within the limits of MIL-STD-461B, Test Method RE02, over the frequency range of 14 KHz to 10 GHz.

3.4.4.4.6 Radiated susceptibility, magnetic induction field, spikes and power frequency. The test set shall not be susceptible to radiated fields in accordance with MIL-STD-461B, Test Method RS02.

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3.4.4.4.7 Radiated susceptibility, electric field, 14 KHz to 10 GHz. The test set shall not be susceptible to radiated field in accordance with MIL-STD-461B, Test Method RS03.

3.4.5 Dimensions. The dimensions of the test set shall be not greater than the following:

a. Instrument Case:

(1) Length	17 inches
(2) Width	15 inches
(3) Height	15 inches

b. Accessory Case:

(1) Length	17 inches
(2) Width	15 inches
(3) Height	15 inches

3.4.6 Weight. Weight of the test set shall be not greater than the following:

a. Instrument Case	50 pounds
b. Accessory Case	40 pounds

3.4.7 Color and finish. Color and finish of the test set shall be in accordance with Drawing 1271AS1000 and MIL-T-28800.

3.4.8 Nameplates and identification marking. Serial number assignment, nameplate approval, and identification marking shall be in accordance with MIL-N-18307.

3.4.8.1 Identification plates. Identification plates for the test set shall be in accordance with MIL-STD-130 and Drawing 1271AS1000.

3.4.8.2 Decals, nameplates and marking locations. All decals, nameplates, and marking shall be affixed to the equipment in accordance with Drawing 1271AS1000 and all referenced marking specifications, documents, and drawings therein.

3.4.8.3 Serial numbers. Serial numbers will be assigned by the procuring activity for the equipment under contract or purchase order. The contractor shall serialize sequentially all equipments which are delivered under contract.

3.5 Performance. The performance of the test set shall be as specified herein in addition to the performance requirements specified in Drawing 1271AS1000. Performance details are also provided in NAVAIR 16-30AWM90-1. Unless otherwise specified herein, these requirements shall apply under both standard (see 3.4.4) and extreme service conditions (see 3.4.4.1). When reduced performance requirements for service conditions are acceptable, such variations are specified in the test requirement paragraph. Figure 2 shows layout of front panel. Figure 3 is a functional block diagram.

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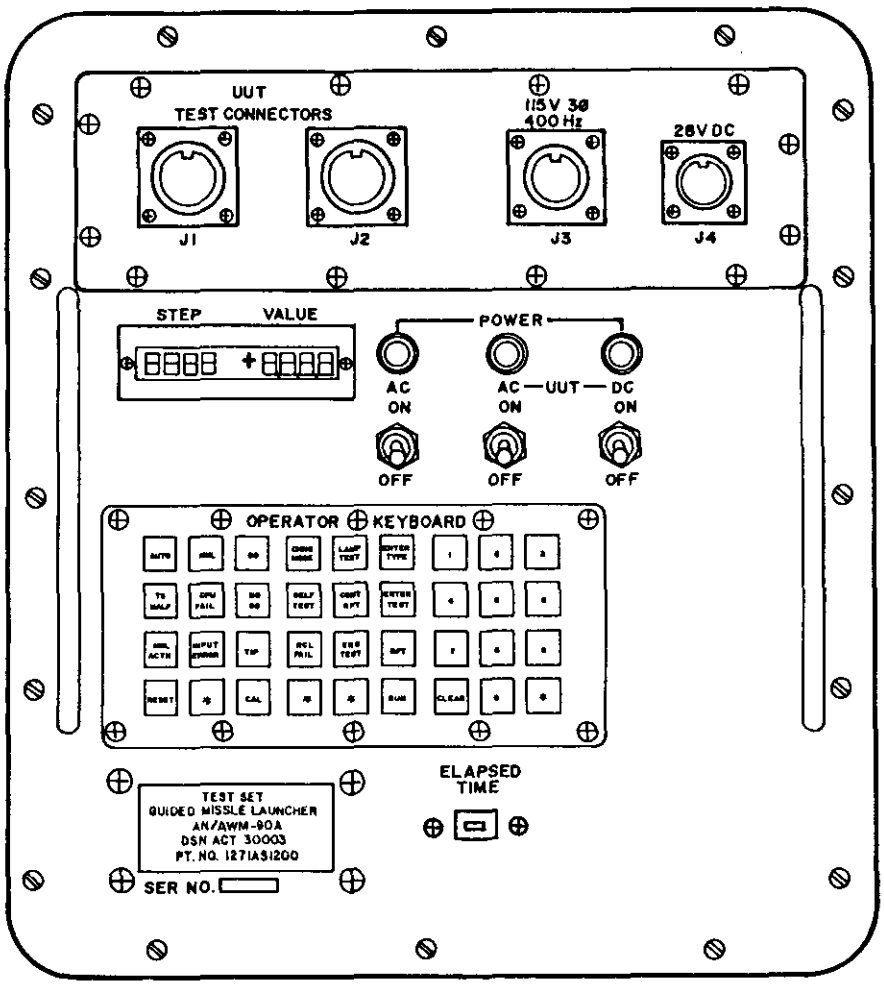


FIGURE 2. Front panel displaying controls indicators and connectors.

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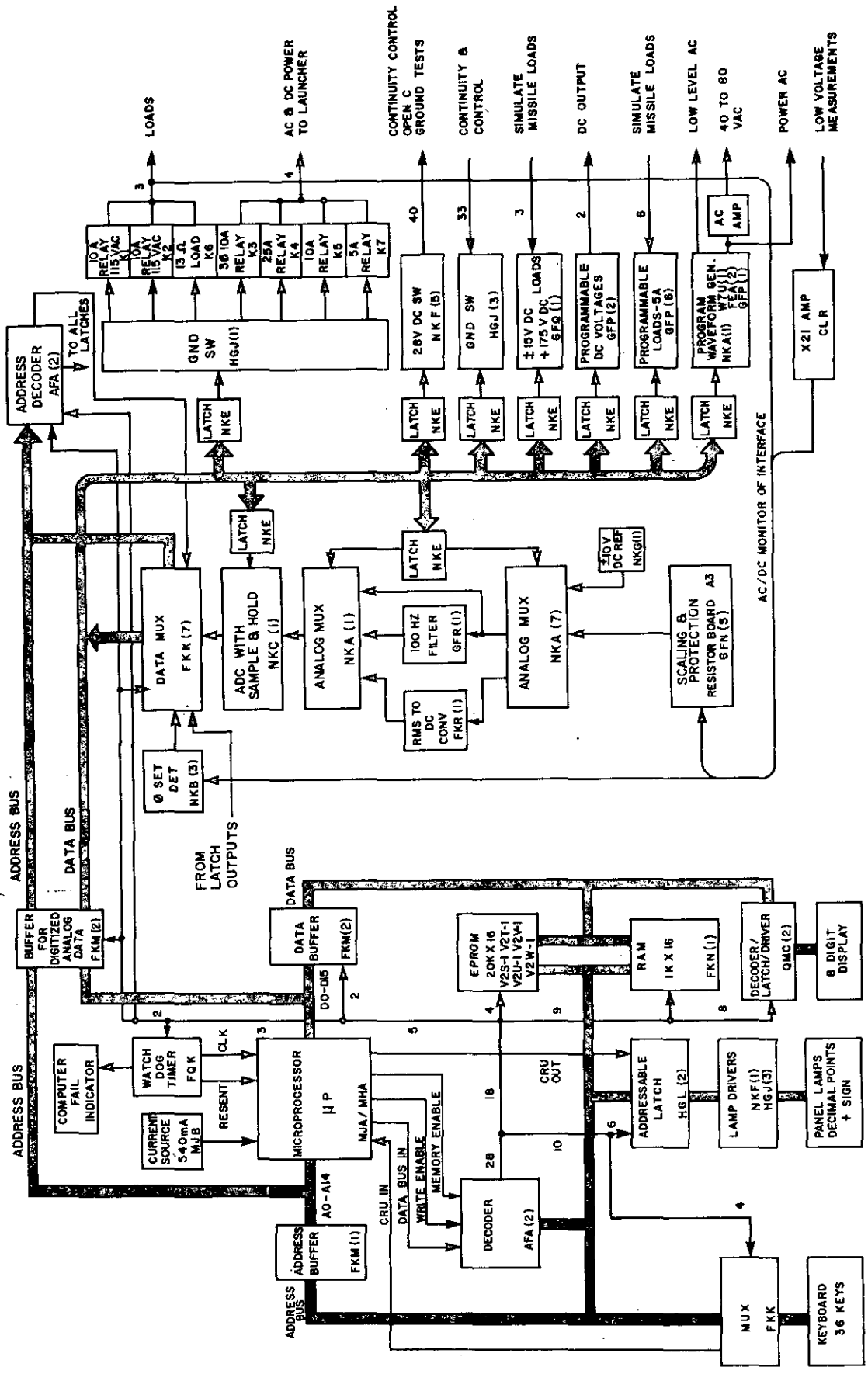


FIGURE 3. Test set block diagram.

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3.5.1 General operation. The test set shall contain a microprocessor which shall control the internal test set operations and flow of execution. The test set, acting as the aircraft interface, shall provide programmed stimuli, stored in the test set's memory and shall measure the response of the equipment being tested, acting as the weapon load. The test set shall then measure the responses of the equipment being tested, shall convert the analog response to digital information by an analog-to-digital converter and shall perform an evaluation upon this information. Results of the evaluation shall be stored in the memory for later references.

3.5.1.1 Function. The test set shall operate in two modes, Automatic and Manual. The Automatic Mode shall be designed to execute the entire set of tests for a particular equipment being tested automatically from start to finish. The Manual mode shall be designed to execute one group of tests at a time as entered by the operator (see 3.5.2.2).

3.5.1.2 Front panel display. The front panel display shall consist of eight seven-segment digits three decimal points and a plus-minus sign. The display shall be divided into two sections. The left section shall contain four digits for indicating the step. The right section shall contain four digits, three decimal points, and a plus-minus sign for indicating the value.

3.5.1.2.1 Step. The step section of the display shall indicate the test that is being executed or if in Recall Failure (RCL FAIL), the test step which failed.

3.5.1.2.2 Value. The value section of the display shall indicate the corresponding test step value, the total number of errors which occurred at end of Automatic testing, a numeric entry as it is made, and TS-MALF and MNL-ACTN codes. Three decimal points and a plus-minus sign shall be available with a value indication.

3.5.1.3 Keyboard. The operator keyboard (see figure 2) shall contain 36 key switches which will light to indicate a condition or an operational option available as an input. The keyboard shall be divided into three sections with each consisting of 12 keys.

3.5.1.3.1 Keyboard section 1 (left). Section 1 of the keyboard shall contain all indicators to inform the operator of the test set's status and the RESET key. The status of operation shall be indicated by the light of the appropriate indicators. Table I lists the indicators and the function of each.

TABLE I. Keyboard section 1 indicators/switches

Key Legend	Key Legend Defined	Function
AUTO	Automatic mode	Indicates that the test set is operating in the automatic mode.
MNL	Manual mode	Indicates that the test set is operating in the manual mode.

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TABLE I. Keyboard section 1 indicators/switches - Continued

Key Legend	Key Legend Defined	Function
GO	Passed test	Indicates that the equipment being tested has passed all executed tests.
TS MALF	Test Set Malfunction	Indicates that a test set malfunction occurred and testing was aborted. A self test should be performed.
CPU FAIL	Computer Failure	Indicates that a failure in the microcomputer occurred. If this occurs during a self test, it indicates that the microprocessor is most probably at fault.
NO GO	Equipment Failed Step	Indicates that the equipment being tested has failed a test step.
MNL ACTN	Manual action	Indicates to the operator that a manual action is required to continue the test operation.
INPUT ERROR	Invalid Entry	Indicates to the operator that an invalid entry was attempted.
TIP	Test in Process	Indicates that an equipment test is in process.
CAL	Calibration mode	Indicates that the test set is in calibration mode (cal box is attached to the test set)
RESET	Reset	This function is always available to the operator. The operator may press this switch any time to reset the test set.
*	Not used	

3.5.1.3.2 Keyboard section 2 (center). Section 2 of the keyboard shall combine indicators and key switches. During different phases of each mode or in different modes, the options of operation shall be indicated by the lighting of these allowable switches. Table II lists the indicator/switches and the function of each.

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TABLE II. Keyboard section 2 indicators/switches

Key Legend	Key Legend Defined	Function
CHNG MODE	Change Mode	Allows the operator to change the mode of operation between Manual and Automatic.
LAMP TEST	Lamp Test	Allows the operator to perform a test which lights all lamps, indicators, and displays. Operator observance required for verification.
ENTER TYPE	Type of Equipment being tested	Allows the operator to enter a number code which represents the equipment type to be tested.
SELF TEST	Self Test	Allows the operator to perform the self test.
CONT RPT	Continuously Repeat a Test	Allows the operator to continuously repeat or execute a particular test. This function is available in the Manual mode only.
ENTER TEST	Enter test	Allows a particular test step to be entered into test set memory for execution. This function is available in the Manual mode only.
RCL FAIL	Recall Failures	At the end of test execution, the number of occurring failures appear in the Value section of the display. RCL FAIL allows the first 25 of these failures to be examined, giving test step and measured value information. Each successive pressing of this switch will cause the next failure information to be displayed. This continues until the last stored failure is displayed. When the next pressing of the switch displays the number of failures, the operator can then step through the failures again.
END TEST		Only used in Manual or Calibration Modes to terminate testing when using Continuous Repeat or performing a test which requires the operator to perform measurements using devices external to the test set.
RPT	Repeat test	Repeats the last test entered. This function is available in the Manual mode only.
RUN	Start test	Execute.
*	Not used.	

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3.5.1.3.2 Keyboard section 3 (right). Section 3 of the keyboard shall be a number key pad for entering number codes, the launcher identification and test steps to be executed. These keys provide the following functions:

a. 0-9. When pressed these keys shall enter the numeric value of that key into the test set's memory. Entry shall be displayed in the Value section of the display as it is entered.

b. CLEAR. When pressed, this key clears the numeric code entered from test set memory.

3.5.2 Test modes. The test set shall be capable of operating in two modes, Automatic and Manual.

3.5.2.1 Automatic. In the Automatic mode, the test set shall service switch inputs until change mode is entered and control is transferred. Under test operations in Automatic mode, the test set shall execute the entire list of tests for equipment under test until completion, unless a test set malfunction or reset occurs. In this mode, the number of NO-GOs which occur shall be counted and the measured value information shall be stored for the first 25 occurring NO-GOs. This information shall then be accessible by pressing RCL FAIL. The first failure step and associated value shall appear in the display. To access any additional failures, each additional pressing of the RCL FAIL key shall cause the next failed step and value to be displayed. After the twenty-fifth failure is displayed, the test set shall return to the beginning of the list and display the total number of failures. During the time that the test set is under Automatic control, the AUTO indicator in section 1 of the keyboard shall be illuminated.

3.5.2.1.1 Switch executions (automatic mode). Switches available for operator's use shall be illuminated in the Automatic mode. The illuminated switches shall be as follows:

a. RUN. After entering the UUT code and the proper cables are connected, the RUN key shall be illuminated. Pressing the RUN key shall cause the test set to execute the list of tests for the UUT.

b. LAMP TEST. Pressing the LAMP TEST key shall cause the test set to execute the lamp test. Operator observance required. (See 3.4.2.3)

c. CHNG MODE. Pressing the CHNG MODE key shall transfer the mode of operation from Automatic to Manual. After pressing the CHNG MODE key, the AUTO indicator in section 1 of the keyboard shall turn off and the MNL indicator shall turn on.

3.5.2.2 Manual. The Manual mode shall be accessed through the Automatic mode and shall be indicated by the MNL indicator in section 1. Under Manual control, the test set shall execute one block of tests. If the operator enters a block number, the test set shall indicate a GO or NO-GO status for that block. If any NO-GOs are encountered, the number of encountered NO-GOs in that block shall be displayed and may be examined by use of the RCL FAIL key as in automatic mode, only the step and measured value information for the first 25 failures are stored. If the operator so chooses, a particular step

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number within a block may be entered and GO/NO-GO and value information for that particular step shall be displayed. The repeat and continuous repeat functions may be used during any of the above operations. The test set shall remain in the Manual mode until a CHNG MODE key is pressed. When CHNG MODE key is pressed, the test set shall return to Automatic mode.

3.5.2.2.1 Switch executions (manual mode). The following switches (keys) shall be used in manual mode:

a. 0-9. Pressing the 0-9 number keys in section 3 shall allow the entry of numeric code for the test step or block desired.

b. CLEAR. This key, when pressed, shall clear the entered number from the test set display to allow another entry.

c. ENTER TEST. Pressing the ENTER TEST key shall cause the test set to examine the entered number to verify that it is a valid test code. Upon verification, the test set shall proceed to execute the test. If the number is invalid, the test set shall light the INPUT ERROR indicator in section 1 and the CLEAR key. TEST SET shall then wait for the CLEAR key to be pressed. Once pressed, the test set shall be ready to accept another command or test code.

d. RPT. Pressing the RPT key shall cause the last test step entered to be executed.

e. CONT RPT. Pressing the CONT RPT shall cause the test set to continuously repeat execution of the last entered test step.

f. END TEST. Only used in Manual or Calibration Modes to terminate testing when using Continuous Repeat or performing a test which requires the operator to perform measurements using devices external to the test set.

g. RUN. When pressed, this key shall allow single step operation through the test list. Each time that RUN is pressed, the next test block shall be executed.

h. CHNG MODE. Pressing the CHNG MODE key shall transfer operation from the Manual mode to the Automatic mode, and the indicators in section 1 shall change from MNL to AUTO.

i. LAMP TEST. Pressing this key, shall cause the lamp test to be executed (see 3.4.2.3).

3.5.2.2.2 Test codes. The test codes for each UUT shall be arranged in sequences of 100. (Example: Test 1 = 100, Test 2 = 200,...)

3.6 Physical characteristics. The test set shall consist of the following assemblies (see figure 1).

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3.6.1 Panel assembly. The panel assembly shall contain all of the electronics, indicators, and controls for performing the test functions. The panel assembly shall be manufactured in accordance with Drawing 1271AS1203.

3.6.2 Test cable (W1). The test cable shall be used to perform self tests and UUT tests. Test cable (W1) shall be manufactured in accordance with Drawing 1532AS515.

3.6.3 Test cable (W2). The test cable shall be used to perform self tests and UUT tests. Test cable (W2) shall be manufactured in accordance with Drawing 1532AS585.

3.6.4 Power cable (W3). The power cable shall be used to supply 115 volts ac power to the test set. Power cable (W3) shall be manufactured in accordance with Drawing 1271AS507.

3.6.5 Power cable (W4). The power cable shall be used to supply 28 volts dc power to the test set. Power cable (W4) shall be manufactured in accordance with Drawing 1271AS512.

3.6.6 LAU-7/A launcher adapter cable (W5). The W5 adapter cable shall be used with test cable W1 to connect test set to LAU-7/A launcher for launcher tests. The adapter cable W5 shall be manufactured in accordance with Drawing 1271AS520.

3.6.7 LAU-7/A launcher/power supply adapter (W6). The W6 adapter shall be used with test cable W2 to connect test set to LAU-7/A launcher or power supply for launcher or power supply tests. Adapter W6 shall be manufactured in accordance with Drawing 1271AS530.

3.6.8 LAU-7/A power supply adapter (W7). The W7 adapter shall be used with test cable W1 to connect test set to LAU-7/A power supply. The adapter W7 shall be manufactured in accordance with Drawing 1271AS590.

3.6.9 LAU-117A(V)2/A launcher adapter (W8). The W8 adapter shall be used with test cable W1 to connect test set to the LAU-117(V)2/A launcher for launcher tests. Adapter W8 shall be manufactured in accordance with Drawing 1271AS540.

3.6.10 LAU-117A(V)2/A launcher adapter cable (W9). The W9 adapter cable shall be used with test cable W2 to connect the test set to the LAU-117A(V)2/A launcher for launcher tests. Adapter cable W9 shall be manufactured in accordance with Drawing 1271AS550.

3.6.11 AERO-5B, LAU-118A(V)1/A launcher adapter cable (W10). The W10 adapter cable shall be used with test cable W1 to connect the test set to the AERO-5B or LAU-118A(V)1/A launcher. Adapter cable W10 shall be manufactured in accordance with Drawing 1271AS560.

3.6.12 AERO-5B, LAU-118A(V)1/A launcher adapter (W11). The W11 adapter shall be used with test cable W2 to connect the test set to the AERO-5B or LAU-118A(V)1/A launcher. Adapter W11 shall be manufactured in accordance with Drawing 1271AS570.

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3.6.13 LAU-115/A (AIM-7F) pylon adapter (W14). The W14 adapter shall be used with test cable W1 to connect the test set to the LAU-115/A pylon. Adapter W14 shall be manufactured in accordance with Drawing 1271AS310.

3.6.14 LAU-116/A (AIM-7F) pylon adapter cable (W15). The W15 adapter cable shall be used with test cable W1 to connect the test set to the LAU-116/A pylon. Adapter cable W15 shall be manufactured in accordance with Drawing 1271AS320.

3.6.15 LAU-115/A, LAU-116/A (AIM-7F) umbilical adapter cable (W16). The W16 adapter cable shall be used with test cable W2 to connect the test set to the LAU-115/A or the LAU-116/A umbilical. Adapter cable W16 shall be manufactured in accordance with Drawing 1271AS330.

3.6.16 CCU cartridge adapter cable (W17). The W17 adapter cable shall be used to establish continuity between the cartridge breeches on the LAU-116/A launcher. Adapter cable W17 shall be manufactured in accordance with Drawing 1271AS340.

3.6.17 LAU-115/A (AIM-9L) pylon adapter (W18). The W18 adapter shall be used with test cable W1 to connect the test set to the LAU-115/A pylon. Adapter W18 shall be manufactured in accordance with Drawing 1271AS350.

3.6.18 LAU-115/A (AIM-9L) umbilical adapter (W19). The W19 adapter shall be used with test cable W2 to connect the test set to the LAU-115/A umbilical. Adapter W19 shall be manufactured in accordance with Drawing 1271AS360.

3.6.19 Self test adapter (W23). The W23 adapter shall be used with cables W1 and W2 to perform self test on test set. Adapter W23 shall be manufactured in accordance with Drawing 1271AS410.

3.6.20 Detent wrench. The detent wrench shall be used in testing the LAU-7/A Launcher. The detent wrench shall be manufactured in accordance with Drawing 67A274C391.

3.6.21 Instrument case and cover. The instrument case shall contain the electronics chassis/front panel. All components shall be removed from the case as an assembly when the front panel is removed. The case and cover shall be manufactured in accordance with Drawing 1271AS1202.

3.6.22 Accessory case and cover. The accessory case shall contain the cables and adapters. The case and cover shall be manufactured in accordance with Drawing 1271AS1502.

3.7 Safety. The contractor shall ensure that safety requirements are in accordance with MIL-T-28800.

3.7.1 Human performance/human engineering. The test set and all components parts thereof, shall be designed in accordance with the

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MIL-STD-1472. Recognized principles of human engineering as specified in MIL-H-46855 shall be followed during design.

3.8 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9 and Drawing 1271AS1000.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)

b. Quality conformance inspection (see 4.5)

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in 3.4.4.

4.4 First article inspection. First article inspection shall be made on one unit representative of the production equipments to be supplied under the contract. First article inspection shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedures (see 4.11). Column I of table III lists minimum tests required for first article inspection. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory acceptable to the Government. No first article inspections shall be conducted prior to approval of first article test procedure by the procuring activity (see 6.2.2).

4.4.1 First article test data. The contractor shall make available all data collected in conducting these tests for procuring activity review and approval (see 6.2.2). This data including a list of all electrical and electronic parts giving their specified voltage, and ambient and surface temperatures shall be made available on first production contracts. The ambient and surface temperatures shall be obtained under the extreme high temperature operating condition.

4.4.2 Scope of first article tests. First article tests shall include all tests listed in table III, column I and any additional tests deemed

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TABLE III. Test requirements

Requirement	I. First Article Inspection	II. Quality Conformance Inspections			
		Individual (A)	Sampling (B)	Special (C) 1/	Life (D)
Visual Examination (3.3, 3.4, 3.4.5 thru 3.4.8.3, and 3.8)	4.5.1.1	4.5.1.1	4.5.1.1	4.5.1.1	4.5.1.1
Performance Characteristics (3.5)	4.5.1.2	4.5.1.2	4.5.1.2	4.5.1.2	4.5.1.2
Workmanship (3.8)	4.5.1.3	4.5.1.3	4.5.1.3	4.5.1.3	4.5.1.3
Environmental (3.4.4.1)	4.7				(See 4.7.1, 4.7.2 and 4.7.3)
Warm-up time (3.4.4.2)	4.8				
Power protection (3.4.4.3.1)		1/	2/	3/	
Interference Control (3.4.4.4)	4.10				
Reliability (3.4.1)	4.5.3		4.5.3.2		
Maintainability (3.4.2)	4.5.4		2/		

1/ Additional individual tests may be requested by the procuring activity (see 4.5.1 and 6.2.1).

2/ Sampling tests to be performed will be determined by the procuring activity based on first article and individual test data (see 4.5.2.1).

3/ Special tests to be performed will be determined by the procuring activity (see 4.5.5.2).

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necessary by the procuring activity to determine that the equipment meets all the requirements of this specification.

4.4.3 First article approval. Approval of first article sample shall be by the procuring activity upon satisfactory completion of all tests listed in table III column I and the first article test report being made available (see 6.2.2).

4.5 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance inspection (see 6.2.2). All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. the contractor shall prepare test reports showing quantitative results for all quality conformance tests (see 6.2.2). Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Quality conformance inspection shall consist of the following tests:

a. Individual Tests (see 4.5.1)

b. Sampling Tests (see 4.5.2)

c. Special Tests (see 4.5.5)

d. Life Tests (see 4.6)

4.5.1 Individual tests. Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

a. Examination of Products

b. Operational Test

c. Manufacturing Run-in Test

4.5.1.1 Examination of product. Each equipment shall be examined carefully to determine that the material and workmanship requirements have been met. Attention shall be given, but not limited to the following:

a. Completeness

b. Nameplates, identification markings and labels

c. Finishes

d. Welded and soldered joints

e. Fit of components

f. Mounting of components

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g. Safety features and interlocks

h. Fastening and securing of devices and parts

i. Cables including plugs and receptacles

j. Grounding connections

k. Weight

l. Overall dimensions

m. Potential shorts and defective insulation

n. Foreign objects, lint, dust

o. Other visual defects

4.5.1.2 Operational test. Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure the equipment operation meets the requirements of this specification.

4.5.1.3 Manufacturing run-in test. Each equipment shall be operated under the conditions specified herein for a period of six hours without failure. A failure shall be defined as anything which causes malfunctioning of the equipment.

a.	Temperature	Ambient room
b.	Humidity	Ambient room
c.	Vibration	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ±3g's

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the six-hour period of operation. Where feasible, the equipment shall be operated during this vibration for the purpose of detecting flaws and imperfect workmanship. The direction of vibration should be vertical to the normal mounting plane for five minutes and lateral to that plane for five minutes. Where it is not feasible to vibrate the equipment in two directions the vertical direction shall be used. During the six-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except that the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration.

4.5.2 Sampling tests. Equipments selected for sampling tests shall first have passed the individual tests. Equipments will be selected for sampling tests by the government inspector in accordance with Table IV.

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4.5.2.1 Scope of sampling tests. As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational test at certain environmental conditions. The conditions may vary for each equipment tested and shall be based on results of the first article, individual, and special tests.

c. Manufacturing run-in test specified in 4.5.1.3 except that the test duration shall be 120 hours with no restriction on the number of failures. However, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipments.

4.5.3 Reliability assurance tests. Reliability assurance tests shall be conducted in accordance with MIL-STD-781. Tests as required by both the qualification phase and the production acceptance (sampling) phase shall be conducted.

TABLE IV. Sampling tests sample selection

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 10	1
Next 50	1
Next 75	1
Next 100	1 for each additional 200 or fraction thereof

4.5.3.1 Qualification phase. Prior to the acceptance of equipments under the contract or purchase order, a minimum of three equipments shall be tested in accordance with MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests". The maximum number of equipments to be used shall be those listed in table V of MIL-STD-781. For the qualification phase, Test Level E shall be used. The accept-reject criteria for Test Plan IV shall be used.

4.5.3.2 Reliability sampling phase tests. Samples of the equipment shall be tested in accordance with MIL-STD-781 under the section entitled "Sampling Phase of Production Reliability Tests". For the sampling phase, Test Level E shall be used, except that the equipment shall operate continuously. The accept-reject criteria for Test Plan V shall be used to determine the length of the tests (until an accept or reject decision is reached).

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4.5.3.2.1 Lot size for sampling phase. The equipments constructed during a two month period shall be defined as one lot. The accept-reject criteria for Test Plan IV shall be used to determine the length of the tests in each lot. Unless otherwise specified in the contract or purchase order, no equipments shall be shipped until an accept decision is reached under each lot (see 6.2.1).

4.5.3.3 Test details. The test details such as the length of the test cycle, the length of the heat portion of the cycle, the performance characteristics to be measured, special failure criteria, and preventive maintenance to be allowed during the test, shall be part of the test procedures to be made available and approved by the procuring activity prior to the beginning of the qualification test phase of the reliability assurance tests (see 6.2.2).

4.5.4 Maintainability test. The Mean-Corrective-Maintenance-Time and Maximum-Corrective-Maintenance Time as specified in 3.4.2 shall be demonstrated as specified in MIL-STD-471, Test Method 9.

4.5.5 Special tests. Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.5.5.1 Special tests schedule. Selection of equipments for special tests shall be made as follows:

a. On an early equipment, after an engineering or material change.

b. Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.5.5.2 Scope of special tests. Special tests shall consist of such tests acceptable to the procuring activity. Test procedures previously approved for the first article tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure for procuring activity approval prior to conducting the tests (see 6.2.2).

4.5.6 Equipment failure. Should a failure occur during either the sampling or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or design defect.

c. Prepare and make available proposed corrective action acceptable to the procuring activity, intended to reduce the possibility of the same failure(s) occurring in future tests (see 6.2.2).

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d. Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.6 Life test. The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300-hours duration and shall be conducted on equipments that have passed the individual tests. The life test shall be performed under the conditions specified in 4.6.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the Sampling Tests, Reliability Tests, or Special Tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed in table V receive at least 300 hours each of test time.

TABLE V. Life test sample selection

Quantity of Equipments Offered for Acceptance	Quantity to be Selected for Sampling Test
First 25	1
Next 175	1
Next 300	1
	1 for each additional 500 or fraction thereof

4.6.1 Test conditions. The life test shall be conducted under the standard conditions specified in 3.4.4.

4.6.2 Test periods. The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.6.3 Performance check. At approximately eight-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be acceptable to the procuring activity (see 6.2.2).

4.6.4 Test data. The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test (see 6.2.2). This record shall indicate the following:

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a. Part type number

b. The circuit reference symbol number

c. The part function

d. Name of the manufacturer

e. Nature of the failure

f. The number of hours which the part operated prior to failure

4.6.4.1 Failure report. In the event of a failure, the government inspector shall be notified immediately. A report shall be prepared and made availble to the procuring activity upon completion of the test (see 6.2.2). In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred (see 6.2.2). The procuring activity will review such proposals and determine whether they are acceptable.

4.7 Environmental tests. The equipment shall be subjected to environmental tests specified in the following paragraph:

4.7.1 High temperature. The test set shall be tested in accordance with MIL-STD-810, Method 501, Procedure I modified as follows:

a. Step 1. Perform operation tests of 4.5.1.2 under the closely controlled conditions of MIL-STD-810.

b. Step 2. The high storage temperature shall be 85°C. The storage temperature shall be maintained for 24 hours. The storage temperature shall be 85°C for four hours and 75°C for the following 20 hours.

c. Step 4. The highest continuous operating temperature shall be 55°C. The test set shall be operated for 20-minutes intermittently at 71°C after the required warmup time. The relative humidity shall be between 20 and 35 percent.

d. Step 5. Perform operation tests of 4.5.1.2.

e. Step 6. The test set shall be operated during the return to standard ambient conditions.

f. The rate of temperature change (steps 2, 4, and 6) shall be between 1° and 5°C per minute.

4.7.2 Low temperature. The test set shall be subjected to low temperature tests in accordance with MIL-STD-810, Method 502, Procedure I modified as follows:

a. Step 1. Perform the operation tests of 4.5.1.2 under the closely controlled conditions of MIL-STD-810.

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b. Step 2. The low storage temperature shall be -62°C. The relative humidity shall be uncontrolled during temperature transition and while at storage temperature. The storage temperature shall be maintained for 24 hours. The test set shall be maintained at -62°C for four hours and then -55°C for the following 20 hours.

c. Step 4. The lowest operating temperature shall be -40°C. Stabilize the test set temperature for two hours.

d. Step 5. Energize the test set and allow the warmup time specified in 3.4.4.2.

e. Step 6. The test set shall be operated during the return to controlled ambient conditions. Stabilize the test set temperatures for two hours.

f. Step 7. Perform the operation test of 4.5.1.2.

g. The rate of temperature change (steps 2, 4, and 6) shall be between 1° and 5°C per minute.

4.7.3 Humidity. The test set shall be subjected to the humidity test specified in MIL-STD-810, Method 507, Procedure III.

4.7.4 Watertight. The test set shall be subjected to a submergence test for watertight equipment in accordance with MIL-STD-108. The test set shall have the case cover latched in place, all drains and ports closed, and the equipment installed prior to performing the test. The test set shall be submerged to a depth that provides a minimum of 0.9m (3 feet) of water above the top of the enclosure.

4.7.5 Altitude. The test set shall be subjected to an altitude test in accordance with MIL-STD-810, Method 500, Procedure I, except the simulated altitudes shall be 15,000 feet (operating) and 40,000 feet (nonoperating). The following details shall modify the MIL-STD-810 when necessary:

a. Pretest data: Perform the operation test of 4.5.1.2.

b. Length of time required for operation and measurements: Warm-up or stabilization time as required in 3.4.4.2, plus a minimum of two hours operation in the operation test of 4.5.1.2.

c. Perform the operation test after return to standard ambient conditions. Degradation of test set performance beyond the specified requirements of 4.5.1.2 shall constitute a failure.

4.7.6 Vibration. The test set shall be subjected to a vibration test in accordance with MIL-STD-810, Method 514, Procedure X, modified as follows:

a. The test set shall be attached directly to the vibration table or to an intermediate structure which is so designed as to be capable of transmitting the specified magnitudes of vibration to the points of the test set attachment throughout the required test frequency range.

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b. During vibration the test set shall be operating in the specified operation check mode.

c. The frequency and displacement shall be as specified in table VI.

d. Linear sweep may be substituted for logarithmic sweep with a sweep time per increment as specified in table VI. Sweep times shall be from low frequency to high frequency to low frequency.

e. A resonance search and resonance dwell shall be conducted as follows:

(1) Resonance search. Resonant frequencies of the test set shall be determined by varying the frequency of applied vibration slowly through the range of 5 to 55 Hz at reduced test levels but with sufficient amplitude to excite the resonance frequencies. Individual resonance search may be performed using the test level and cycling time specified for the cycling test, provided the resonance search time is included in the required cycling test time.

(2) Resonance dwell. The test set shall be vibrated along each axis at the most severe resonant frequency determined in the resonance search for that axis. Test levels, frequency ranges, and test times shall be in accordance with table VI. If a change in the resonance frequency occurs during the test, its time of occurrence shall be recorded and immediately the frequency shall be adjusted to maintain the peak resonance condition. The final resonant frequency shall be recorded. If no significant resonant response is found, the test set shall be vibrated for 10 minutes (resonance dwell) at 33 Hz.

f. Perform the operation tests required in 4.5.1.2.

TABLE VI. Vibration test points

Frequency Increments Hz	Vibratory Level (Inches peak-to-peak)	Sweep Time per 1/ Increment (Minutes)	Survey Time per axis (Minutes)	Dwell Time per axis (Minutes)
5 to 15 15 to 25 25 to 55	1.5mm (0.06 inch) 1.0mm (0.04 inch) 0.5mm (0.02 inch)	7 3 5	15	10

1/ Sweep times specified shall be for a low frequency to high frequency to low frequency variation for the applicable increment.

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4.7.7 Shock. The test set shall be subjected to a shock test in accordance with MIL-STD-810, Method 516, Procedure I, using the half-sine shock waveform of the Half Sine Shock Pulse Configuration and its Tolerance Limits figure of MIL-STD-810, nominal duration (D) of 11 ms. The test set shall be given three shocks in each direction and each axis (total 18). Peak value (A) of each shock shall be 30 g's. The test set shall be operating during the shock. After the shock exposure, perform the operation test specified in 4.5.1.2.

4.7.8 Transit drop. The test set with contained test equipment shall be subjected to a transit drop test in accordance with MIL-STD-810, Method 516, Procedure II, using a drop height of 760mm (30 inches). The test set shall be dropped once on each corner and each face for a total of 14 drops. Perform the operation test specified in 4.5.1.2.

4.7.9 Bench handling. The test set shall be subjected to bench handling test of MIL-STD-810, Method 516, Procedure V, modified as follows:

a. The test set shall be tested while in the case and operating in the operation test of 4.5.1.2.

b. The test set shall be tested removed from the case nonoperating.

c. Examine the instrument for mechanical damage. Damage to any control, indicator, fuseholder, connector, or other protruding component part shall constitute a failure.

d. Perform the operation test of 4.5.1.2.

4.8 Warm-up time. The equipment shall be subjected to tests to determine the capability of the equipment to conform to the warm-up time requirements of 3.4.4.2.

4.9 Input power test.

4.9.1 Voltage and frequency variation tests. The equipment response to the input voltage and frequency variations shall be tested in accordance with MIL-T-28800. Operational performance may be degraded during the exposure but equipment shall not sustain damage and shall return to normal operation when test is completed.

4.9.2 Voltage and frequency-transient test. The equipment performance during voltage transients shall be tested in accordance with MIL-T-28800. Operational performance may be degraded during the exposure but equipment shall not sustain damage and shall return to normal operation when test is completed.

4.10 Electromagnetic interference (EMI) tests. EMI tests shall be performed in accordance with MIL-STD-462 to determine that the equipment meets the requirements of MIL-STD-461B for Class B equipment (see 3.4.4.4). The test set shall be operating during the EMI tests. The test set display shall be observed for signs of vulnerability during susceptibility tests.

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4.10.1 Conducted emissions, power and interconnecting leads, 0.015 to 50 MHz. The test set shall be subjected to a conducted emission test on the power and interconnecting leads in accordance with MIL-STD-462, Method CE03.

4.10.2 Conducted emissions, power leads, spikes, time domain. The test set shall be subjected to a conducted emission test on the power leads in accordance with MIL-STD-462, Method CE07.

4.10.3 Conducted susceptibility power leads, 30 Hz to 50 KHz. The test set shall be subjected to signals between 30 Hz to 50 KHz being injected on the dc power leads in accordance with MIL-STD-462, Method CS01.

4.10.4 Conducted susceptibility power leads, 0.05 to 400 MHz. The test set shall be subjected to signals between 0.05 to 400 MHz being injected on the dc power leads in accordance with MIL-STD-462, Method CS02.

4.10.5 Conducted susceptibility, spikes, power leads. The test set shall be subjected to the injection of spikes on all ungrounded ac and dc input power leads in accordance with MIL-STD-462, Method CS06.

4.10.6 Radiated emissions, electric field, 14 KHz to 10 GHz. The test set shall be probed for points for maximum radiation over the frequency range of 14 KHz to 10 GHz in accordance with MIL-STD-462, Method RE02.

4.10.7 Radiated susceptibility, magnetic induction field, spikes and power frequency. The test set shall be subjected for short duration induction fields in accordance with MIL-STD-462, Method RS02.

4.10.8 Radiated susceptibility, electric field, 14 KHz to 10 GHz. The test set shall be subjected to an electric field in the frequency range of 14 KHz to 10 GHz in accordance with MIL-STD-462, Method RS03, except the strength of radiated field shall be 10 volts/meter peak in the frequency range of 14 KHz to 30 MHz and five volts/meter peak in the frequency range of 30 MHz to 10 GHz.

4.11 Test procedures. The procedures used for conducting first article tests, quality conformance tests, and life tests shall be prepared by the contractor for procuring activity review and acceptance (see 6.2.2). The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract (see 6.2.1). MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts, such procedures shall be provided and may be used when their use is acceptable to the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures, including additional tests when deemed necessary (see 6.2.1).

4.12 Presubmission testing. No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply with all applicable requirements.

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4.13 Inspection of packaging. The sampling and inspection of the preservation, packing and container marking shall be in accordance with the requirements of Section 5 and the standards and specifications listed therein.

5. PACKAGING.

5.1 Preservation. Preservation shall be level A or industrial as specified (see 6.2.1).

5.1.1 Level A.

5.1.1.1 Cleaning. Test sets shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying. Test sets shall be dried in accordance with MIL-P-116.

5.1.1.3 Unit packs. Unless otherwise specified in the contract, (see 6.2.1), each test set shall be individually unit packed in accordance with the methods of MIL-P-116 designated herein ensuring compliance with the applicable requirements of that specification.

5.1.2 Industrial. The industrial preservation of test sets shall be in accordance with the requirements of MIL-STD-1188.

5.2 Packing. Packing shall be level A or industrial, as specified in the contract (see 6.2.1).

5.2.1 Level A. The packaged test sets shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirement in the Appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joints with tape, two inches minimum width, conforming to PPP-T-60, Class 1 or PPP-T-76. Banding (reinforcement requirements) shall be applied in accordance with the Appendix to PPP-B-636 using nonmetallic or tape banding only.

5.2.2 Industrial. The preserved test sets shall be packed in accordance with the requirements of MIL-STD-1188.

5.2.3 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or purchase order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.3.1 Level A. Test sets, packed as specified in 5.2.1, shall be unitized on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

5.3 Marking.

5.3.1 Level A. In addition to any special or other identification marking specified in the contract (see 6.2.1), each unit pack, intermediate

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and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.3.2 Industrial. Industrial marking shall be in accordance with the requirements of MIL-STD-1188.

5.4 General.

5.4.1 Exterior containers. Exterior containers (see 5.2.1 and 5.2.2) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

5.4.2 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.13.

6. NOTES

6.1 Intended use. The Guided Missile Launcher Test Set AN/AWM-90A is intended for use at Intermediate Level guided missile launcher shops to test AERO 5B, LAU-118(V)1/A, LAU-7/A-5, LAU-115/A, LAU-116/A, LAU-117A(V)2/A Guided Missile Launchers.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Selection of applicable levels of packaging and packing (see 5.1).

c. When a first article is required for inspection and approval (see 4.4).

d. Additional individual tests, if required (see 4.5.1 and table IV).

e. Lot acceptance for each lot, if other than specified herein (see 4.5.3.2.1).

f. Test procedure modifications or additional tests, if required (see 4.11).

g. Unit packs, if other than specified herein (see 5.1.1.3).

6.2.2 Data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423, Contract Data Requirements List (CDRL) the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved CDR incorporated into the contract. When the provisions of DAR 7-104.9 (N) (2) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in

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accordance with the contract or purchase order requirements. Deliverable data required by this specification is cited in the following paragraphs.

Paragraph no.	Data requirement title	Applicable DID no.
4.4, 4.11	Procedure, First Article Inspection	DI-T-4901
4.4.1, 4.4.3	Report, First Article Inspection	DI-T-4902
4.5.5, 4.5.5.2, 4.6.3, 4.11	Procedures, Production/Acceptance Inspection	DI-T-4903
4.5, 4.6.4	Reports, Production Inspection	DI-T-4904
4.5.3.3	Procedures, Reliability Tests	DI-R-7035
4.6.4.1	Engineering Change Proposals (ECPs) and Request for Deviations and Waivers (Long Form)	DI-E-2037
4.5.6c	Plan, Failure Data Collection, Analysis and Corrective Action	UDI-T-23719
4.6.4.1	Report, Failure	UDI-R-21141A

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5000.19L., Vol. II, AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

6.3 First article. When a first article inspection is required, the item will be tested and should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations and test approval of the document's first article.

Preparing Activity
NAVY-AS
(Project 4935-N051)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-T-85641 (AS)	Test Set, Guided Missile Launcher AN/AWM-90 A

3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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